

Computer Science & IT

Discrete Mathematics



Graph Theory

Lecture No. 05



By- Vishal Sir

Recap of Previous Lecture



Topic

Havel Hakimi's algorithm



Topic

Complement of a graph

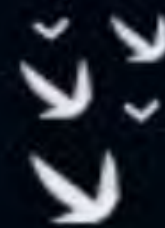


Topic

Graph isomorphism



Topics to be Covered



Topic

Graph isomorphism

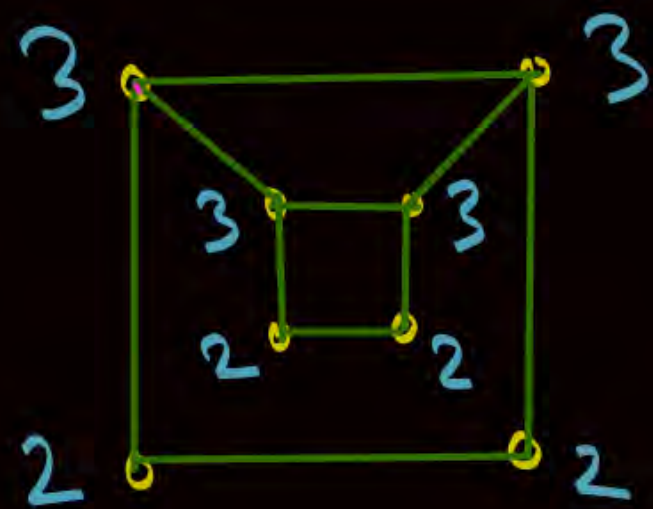


Topic

Self-complementary graph



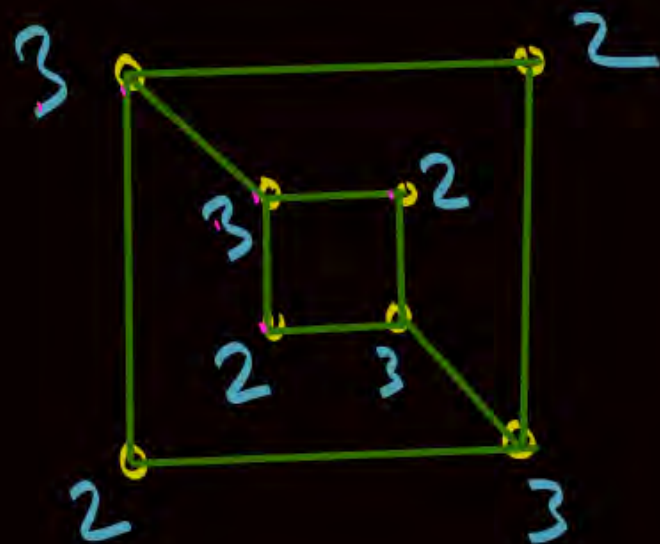
H.W. Q: Consider two graphs G_1 & G_2



G_1

Deg seq. of $G_1 = \{3, 3, 3, 3, 2, 2, 2, 2\}$

Check whether the graphs are isomorphic or not?



G_2

Deg seq. of $G_2 = \{3, 3, 3, 3, 2, 2, 2, 2\}$

✓ ① $|V(G_1)| = |V(G_2)|$

✓ ② $|E(G_1)| = |E(G_2)|$

✓ ③ Deg. seq. of $G_1 = \text{Deg. seq. of } G_2$

✗ ④ The only 4 vertices of deg=3 in G_1 form a cycle of length '4', but vertices of deg=3 in G_2 does not form any cycle

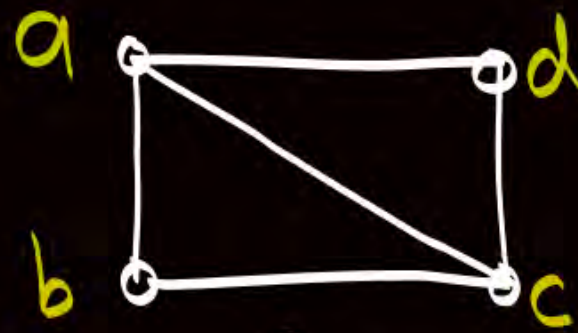
∴ G_1 & G_2 are not isomorphic



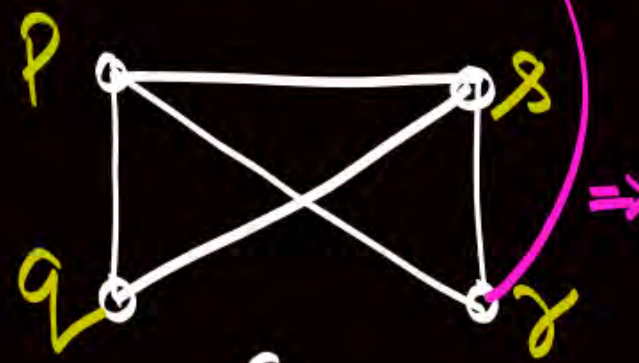
Topic : Graph Isomorphism

Note:- Two simple graphs G_1 and G_2 are isomorphic to each other if and only if their complements are isomorphic to each other

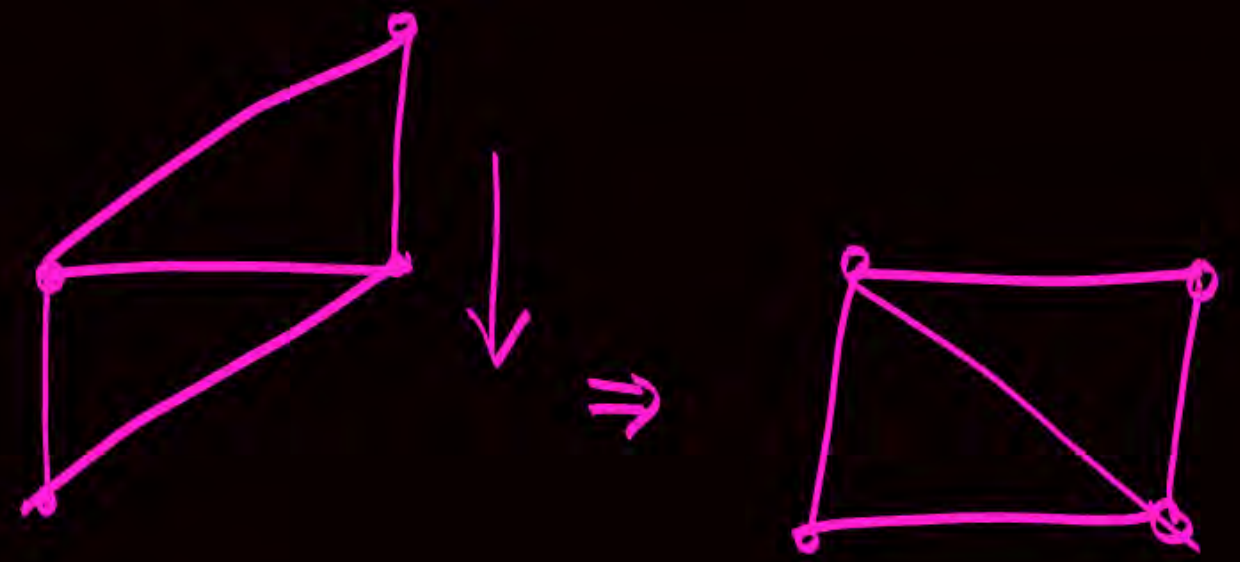
Q: Consider two graphs G_1 & G_2



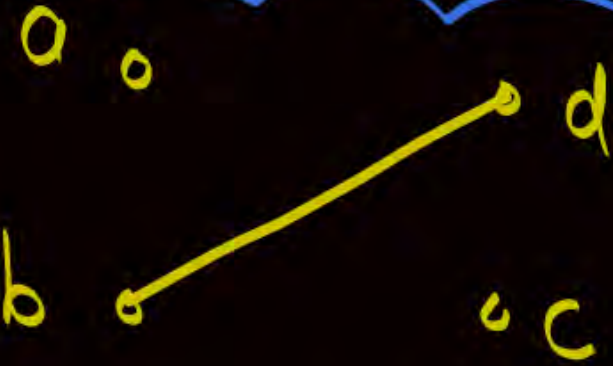
G_1



G_2



Check whether G_1 & G_2 are isomorphic or not?



$\overline{G_1}$



$\overline{G_2}$

We can observe that

$$\overline{G_1} \cong \overline{G_2}$$

$\therefore$ We can say that

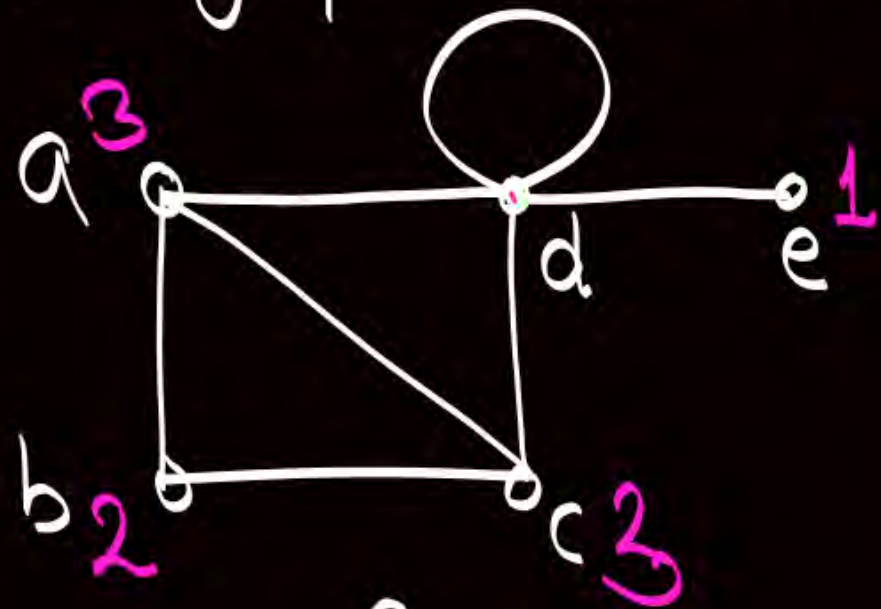
$$\boxed{G_1 \cong G_2} \quad \underline{\underline{\text{Ans}}}$$



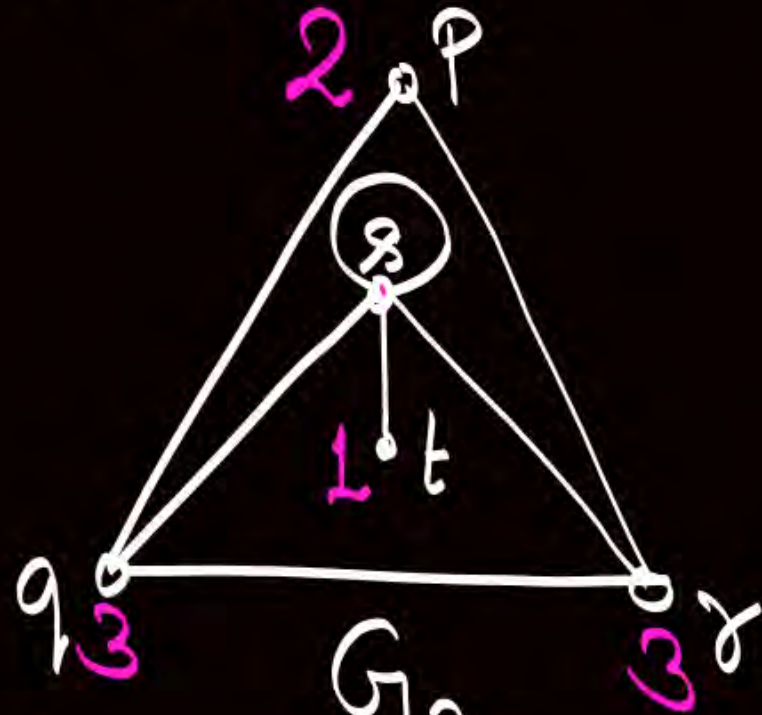
Topic : Graph Isomorphism

- ★ Two graphs G_1 and G_2 are isomorphic if and only if their corresponding subgraphs {i.e the graph obtained by deleting some vertices from graph G_1 and the graph obtained by deleting the the images of those vertices from G_2 } are isomorphic to each other

Q: Consider two graphs G_1 and G_2 , and check whether the graphs are isomorphic or not?



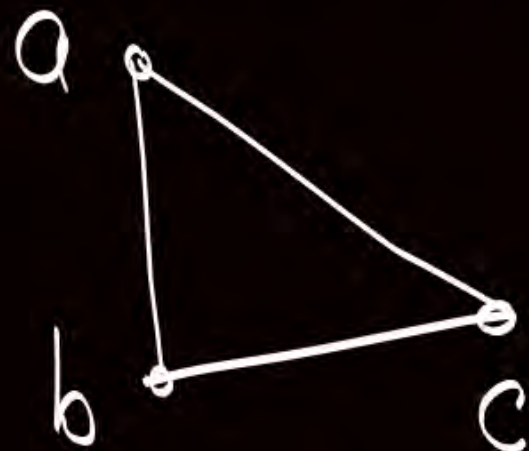
G_1



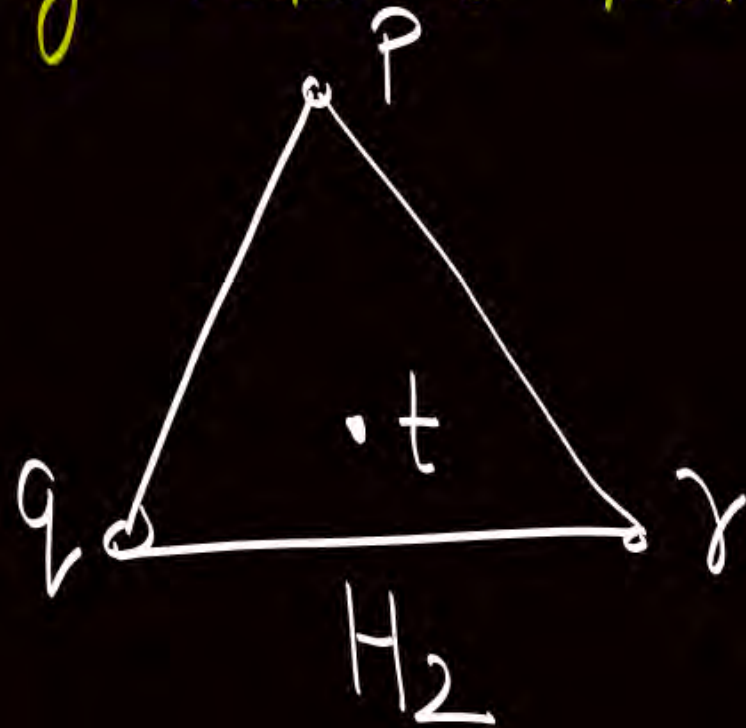
G_2

We can observe that image of vertex 'd' in G_1 is vertex 's' in G_2

Obtain the subgraph by deleting vertex 'd' from G_1 and its image 's' from G_2



H_1



H_2

We can observe

$$H_1 \cong H_2$$

$$\therefore \boxed{G_1 \cong G_2} \text{ Ans}$$

Q:- How many simple 'non-isomorphic' graphs are possible with '3' labeled vertices.

Q:- How many simple 'non-isomorphic' graphs are possible with '4' labeled vertices.

HW Q:- How many simple 'non-isomorphic' graphs are possible with n ($n \geq 4$) labeled vertices and two edges.



2 mins Summary



Topic

Graph isomorphism



Topic

Self-complementary graph

THANK - YOU